

End Sem Lab Exam

Introduction to Programming (CSE 101)- 2024

Duration: 75 mts

Total: 24 Marks

Please read the following instructions carefully:

- **PLEASE NOTE.** You are allowed to have **only your IIITD Id cards, laptops, chargers and pencil box in the exam halls. Having a phone or any other communication device will be considered as using unfair means.** This is an exam - any violation will be reported to the authorities concerned without further warning.
- **Please leave your belongings or bag (with your books, phones, smart watch, etc,) with the security on the ground floor of the LHC.**
- Marks for each question are given.
- Read the questions carefully before attempting the same.
 - In some questions you are asked to use a given function signature. It is mandatory to follow this strictly.
 - You are required to write the test functions wherever it is mentioned explicitly in the question.
- Any clarifications will be given on the screen in the class for all to see.
- Use any editor or IDE to create the .py solution files
- For each question, the code must be saved in a single file named **QNo_RollNo.py**.
- At the end of the exam, zip all the tested .py files into **RollNo.zip** and upload it on the Google Classroom

1. **(6 marks)** Consider a dictionary D which contains 10 key-value pairs with keys as 0, 1, ..., 9. Each key stores a **sorted** list of integers as values. Moreover, the integers (values) in each list have the same leftmost digit (e.g., 1 for 19, 3 for 37, 0 for 6, etc.) as its corresponding key.

For example,

```
D = {
    0: [2, 5, 6],
    1: [11, 17, 19],
    2: [29],
    3: [],
    ...
    9: [93, 98, 99]
}
```

Write a function `FindInDict()` which will receive a dictionary and an integer as parameters and returns (key, index) pairs, where key and index are as follows:

- **If found:** Return the key in the dictionary and index in the corresponding list.
- **If not found:** Insert the integer at the appropriate location (based on the key corresponding to the integer) and return the key and index of the inserted item.

Note:

- For simplicity, we assume positive integers up to 99.
- Single digit numbers should belong to key 0.

The following function signature must be strictly followed. You are free to define additional functions as per requirement.

```
def FindInDict(D, number):  
  
    # Write your code here  
  
    return key, index
```

Sample Input 0:

FindInDict(0: [2, 5, 6], 1: [11, 17, 19], 2: [29], 3: [], 9: [93, 98, 99], 17)

Sample Output 0:

1, 1

Sample Input 1:

FindInDict(0: [2, 5, 6], 1: [11, 17, 19], 2: [29], 3: [], 9: [93, 98, 99], 47)

Sample Output 1:

4, 0

(Here, the dictionary should get updated to {0: [2, 5, 6], 1: [11, 17, 19], 2: [29], 3: [], 4:[47], 9: [93, 98, 99], 47})

Test Function: You must define a function, `_test()`, with at least two assert statements (for two different situations) for testing the `FindInDict()` function.

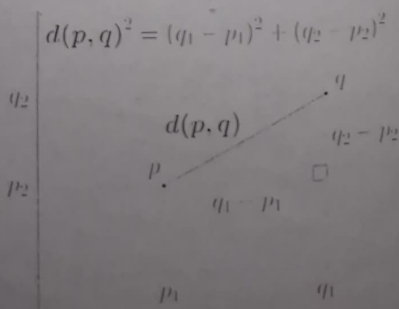
2. (6 marks) Consider a file which contains the coordinates of various points in a 2D space as follows:

(34, 98), (-16, 34), ..., (112, -819),
(-87, -12),
(6, 4),

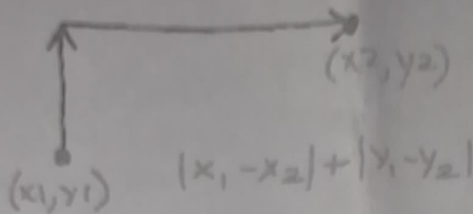
Write a function `nearestPoint()` which receives a filename and two integers (x and y as coordinate of a new point P) as inputs. The function should return the nearest neighbour of the new point P based on the following distance functions.

Formula:

- Euclidean distance (only for students with **EVEN** roll number):



- Manhattan distance (only for students with ODD roll number)



The following function signature must be strictly followed. You are free to define additional functions as per requirement.

```
def nearestPoint(fileName, x, y):

    # Write your code here

    return x_P , y_P # nearest point
```

Sample Input 0:

nearestPoint(input.txt, 2,3) (where input.txt contains the points given above - (34, 98), (-16, 34), ..., (112, -819), (-87, -12), (6, 4))

Sample Output 0:

6, 4

Test Function: You must define a function, `_test()`, with at least two assert statements (for two different situations/conditions) to test the `nearestPoint()` function.

3. (6 marks) Create a Python class named `Fraction` to represent fractions in the form a/b , where a and b are integers, and $b \neq 0$. Your class should support the following operations and functionalities:

- **Add Two Fractions:** Implement a dunder function to add two fractions and return the result as a new `Fraction` object.
- **Multiply Two Fractions:** Implement a dunder function to multiply two fractions and return the result as a new `Fraction` object.
- **Print the fraction number** in the format a/b using the dunder function.
- **Reduce Fractions:** Ensure that every fraction is reduced to its simplest form (i.e., $\gcd(a, b) = 1$) after every operation.

Write a function that uses the above class and methods to perform the following:

- Create an object `frac = Fraction()`
- Use `frac` to reduce a fraction accepted as input from the user.
- Print the result in the reduced form.
- Add another fraction (accepted as input) to the above reduced form. Apply *Reduce Fraction* to the result and print it.
- Multiply the result of the above with another fraction. Apply *Reduce Fraction* to the result and print it.

4. (4 marks) The canteen in the Institute has a menu of items with unit price for each. For your program, the menu is stored as a list of tuples (each tuple giving the item name and the unit price) as:

```
menu = [("Samosa", 15), ("Idli", 30), ("Maggie", 50), ("Dosa", 70), ("Tea", 10),  
("Coffee", 20), ("Sandwich", 35), ("ColdDrink", 25)]
```

Write a program that repeatedly takes a string (item name) and a number (quantity) as input, and when no input is given (i.e., a return is hit), it prints the bill of the items ordered as per the output format given below:

Input:

Menu: [("Samosa", Rs 15/piece), ("Idli", Rs 30/piece), ("Maggie", Rs 50/piece), ("Dosa", Rs 70/piece), ("Tea", Rs 10/piece), ("Coffee", Rs 20/piece), ("Sandwich", Rs 35/piece), ("ColdDrink", Rs 25/piece)]

Please place your order from above menu (Item and quantity): Maggie 1

Want more? Item and quantity please: Samosa 5

Want more? Item and quantity please: #A return is hit without any input.

Output:

Item <tab> Qty <tab> Amount

Maggie <tab> 1 <tab> Rs 50

Samosa <tab> 5 <tab> Rs 75

TOTAL (6 items): Rs 125

5. (4 marks) Write a program to print the following pattern for an integer N using recursion only.

Note: Iterative solutions are not acceptable. Your program should not have any looping statements.

Hint: You may need to define two recursive functions -- one for the row (line) and another for the column.

For N = 5, pyramid should be as follows

```
1  
1 2  
1 2 3  
1 2 3 4  
1 2 3 4 5  
1 2 3 4  
1 2 3  
1 2  
1
```

End